Weekly check

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Find the range of 0, 2, 5, 10.

2. Find the MAD of 2, 5, 5, 8.

3. Data are 1, 2, 4, 4, 5, 6, 6. How many dots belong above 4 in a dot plot?

4. Four scores have mean 7. Three are 4, 5, 9. Find the fourth.

5. A triangle has base 9 cm and height 4 cm. Find its area.

6. Find the distance from (-4, 2) to (3, 2).

7. Write an inequality for x that means no more than 12. Is the boundary included?

8. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

9. How many $\frac{1}{3}$-meter pieces fit in 2 meters?

10. Luis finds the median of 9, 2, 6, 3 by averaging 2 and 6. He gets 4. What should he change?

11. Choose the statistical question: (A) How many pages are in one named book? (B) How many pages are in each of 14 books? Explain.

12. Make a dot plot for 7, 7, 8, 9, 9, 9, 11. State the total dots and most frequent value.

13. Find histogram frequencies for 6, 7, 8, 8, 9, 10, 11, 11 using bins 6 to under 8, 8 to under 10, and 10 to under 12.

14. Find the mean of 8, 10, 12, 14. Explain it as a fair share.

15. Order 14, 8, 11, 10, 11, 15 and find the median.

16. Four values have mean 12. Three are 9, 11, 13. Find the missing value.

17. Find the range of 7, 9, 11, 14, 17. Which values determine it?

18. Find the five-number summary and IQR of 7, 8, 10, 12, 14, 16, 18. Exclude the median from each half.

CHALLENGE / Explain your plan and check the result

C1. Data are 2, 4, 6, 8. Find the mean, median, and MAD. Duplicate every observation so the new data contain eight values. Predict and verify which of the three measures changes.

C2. A data set has five whole-number values with minimum 2, maximum 10, median 6, and mean 6. Give one possible set. Find its range and five-number summary using the exclude-the-median rule. Explain whether your set is the only possible one.